

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TNAS)
ANSWER KEY
PAPER-1
PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	A	C	C	B		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,B,C	D	A,B,C	B,C,D	A,B,C	A,C
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	20.00	10.00	4.00	5.50	10.00	2.50
SECTION-II (ii)	Q.	7	8	9			
	A.	35	225	60			

PART-2 : CHEMISTRY

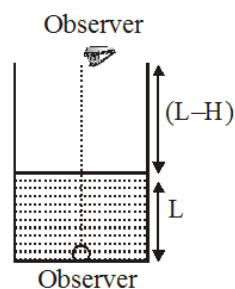
SECTION-I (i)	Q.	1	2	3	4		
	A.	D	D	C	D		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	B,C,D	A,D	A,B,C	A,B,C	A,B,C	B,C,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	10.00	2.00	3.00	2.00	1.50	0.66 to 0.67
SECTION-II (ii)	Q.	7	8	9			
	A.	4	709	4			

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	B	C	C		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,B	A,B	A,B,D	A,B,C,D	A,B,C,D	B,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	0.50	0.25	0.00	1.00	6.00	4.00
SECTION-II (ii)	Q.	7	8	9			
	A.	1	4	3			

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)

1. Ans (A)



$$X_{app} = L - H + \frac{H}{\mu}$$

$$\frac{dX_{app}}{dt} = \frac{dH}{dt} \left(\frac{1-\mu}{\mu} \right)$$

$$\pi r^2 H = V$$

$$\therefore \frac{dH}{dt} = \frac{2H}{r} \frac{dr}{dt} = \frac{2KH}{r}$$

$$\therefore \frac{dX_{app}}{dt} = \frac{2KV}{\pi r^3} \left(\frac{(1-\mu)}{\mu} \right)$$

2. Ans (C)

$$\text{Use } \int_0^4 F dt = mv - m(5)$$

$$\text{work}_F = \frac{1}{2}mv^2 - \frac{1}{2}m(5)^2$$

3. **Ans (C)**

$$F_{\text{ext}} = kx + mg \quad (x : \text{extension at any } t)$$

$$W_{\text{ext}} = \int_0^{x_0} (kx + mg) dx, \text{ where } kx_0 = mg$$

Releasing, total work done = $W_{\text{ext}} + mg(h)$,
where h : natural length

4. **Ans (B)**

$$m \frac{dv}{dt} v = P = \text{Constant}$$

When acceleration reduces to half value its speed becomes doubled i.e. $v = 2v_0$.

$$Pt = \frac{1}{2} m(v^2 - v_0^2)$$

$$Pt = \frac{1}{2} m(4v_0^2 - v_0^2)$$

$$\Rightarrow t = \frac{3mv_0^2}{2P} = \frac{3 \times 1 \times 100 \times 15}{2 \times 1000} = 2.25 \text{ sec}$$

PART-1 : PHYSICS

SECTION-I (ii)

5. **Ans (A,B,C)**

Use $g_{\text{eff}} = g \sin 37 = \frac{3g}{5}$ instead of g in all standard vertical circle cases

6. **Ans (D)**

$$D = \sqrt{\frac{2H}{g}} \sqrt{2g(2R)}$$

7. **Ans (A,B,C)**

$$\text{Impulse by field is } I = qE\Delta t = 10^{-6} \times \frac{\pi}{10} \times 10^6 \times 1 \times 10^{-3}$$

$$\Rightarrow I = \pi \times 10^{-4} \text{ kg m/s}$$

$$\Rightarrow \Delta v = \frac{I}{m} = \frac{\pi}{10} \text{ m/s}$$

$$\text{Also, } \omega = \sqrt{\frac{k}{m}} = \pi \text{ rad/s and } T = 2\text{s.}$$

$$\text{At } t=0, V = \frac{\pi}{10} \text{ m/s} = A\omega, A = 0.1\text{m}$$

at $t = 0.5$ s, block is at extreme but again receives the impulse, gaining velocity $\frac{\pi}{10}$ m/s. This makes the new amplitude as $\sqrt{2} \times 0.1 = 0.1414\text{m}$
(using $v = \omega \sqrt{A^2 - x^2}$)

Also, another

$$\frac{1}{4} \text{ s it will take from } x = 0.1 \text{ m to}$$

$$x = \sqrt{2} \times 0.1 \text{ m. (using}$$

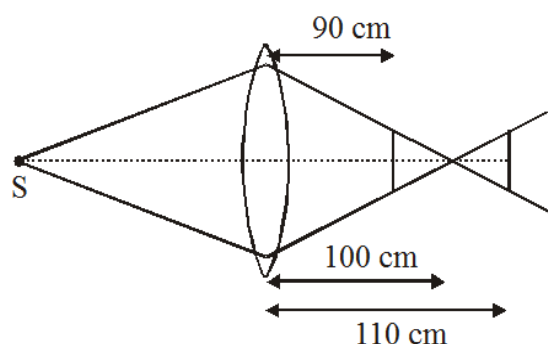
$$x = A \cos \omega t)$$

The particle comes back to mean position at $t = 1.5\text{s}$ when the impulse makes its velocity zero. So, no compression occurs.

8. **Ans (B,C,D)**

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$V = 100 \text{ cm}$$



9. **Ans (A,B,C)**

In figure-1

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} + \frac{1}{-60} = \frac{1}{-30}$$

$$V = -60$$

$$\text{Hence } m = -\frac{v}{u} = -1$$

In figure-2, images will not separated.

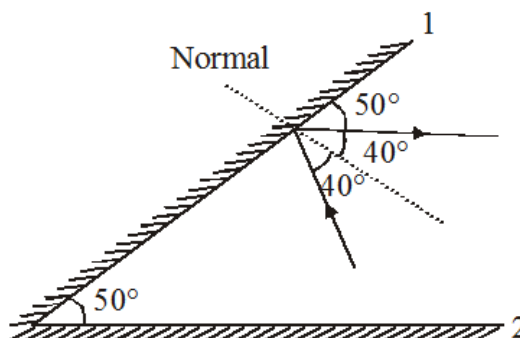
10. **Ans (A,C)**

Deviation

$$\delta_1 = \pi - 2\theta \text{ (Clockwise)}$$

$$\delta_2 = \pi - 2(50 + \theta) \text{ (Anti-clockwise)}$$

$$\delta = \delta_1 - \delta_2$$



PART-1 : PHYSICS

SECTION-II (i)

1. **Ans (20.00)**

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u} = \frac{1}{30} + \frac{1}{60}$$

$$\Rightarrow \frac{1}{f} = \frac{1}{20} \Rightarrow f = 20 \text{ cm}$$

Focal length of convex mirror

$$f = \frac{1}{2} (30 - 10) = 10 \text{ cm}$$

2. **Ans (10.00)**

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u} = \frac{1}{30} + \frac{1}{60}$$

$$\Rightarrow \frac{1}{f} = \frac{1}{20} \Rightarrow f = 20 \text{ cm}$$

Focal length of convex mirror

$$f = \frac{1}{2} (30 - 10) = 10 \text{ cm}$$

3. **Ans (4.00)**

Let elongation in spring A, B and C be x_1 , x_2 and x_3 respectively.

Considering spring forces and constant relations

$$x_2 = 4x_3 \quad \dots(i)$$

$$x_2 = 2x_1 \quad \dots(ii)$$

$$\text{and } x_1 + 2x_2 + x_3 = x \quad \dots(iii)$$

$$\Rightarrow x_1 = \left(\frac{2}{11}\right)x; x_2 = \left(\frac{4}{11}\right)x; x_3 = \left(\frac{1}{11}\right)x$$

$$\text{Also, } F = 2k \left(\frac{x}{11}\right)$$

$$\Rightarrow T = 2\pi \sqrt{\frac{11m}{2k}}$$

4. **Ans (5.50)**

Let elongation in spring A, B and C be x_1 , x_2 and x_3 respectively.

Considering spring forces and constant relations

$$x_2 = 4x_3 \quad \dots(i)$$

$$x_2 = 2x_1 \quad \dots(ii)$$

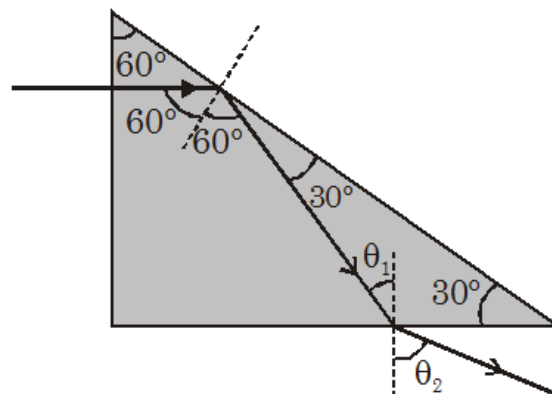
$$\text{and } x_1 + 2x_2 + x_3 = x \quad \dots(iii)$$

$$\Rightarrow x_1 = \left(\frac{2}{11}\right)x; x_2 = \left(\frac{4}{11}\right)x; x_3 = \left(\frac{1}{11}\right)x$$

$$\text{Also, } F = 2k \left(\frac{x}{11}\right)$$

$$\Rightarrow T = 2\pi \sqrt{\frac{11m}{2k}}$$

5. **Ans (10.00)**



$$\mu_1 \sin \theta_1 = \mu_2 \sin \theta_2$$

$$\frac{5}{3} \sin 30^\circ = \frac{4}{3} \sin \theta_2$$

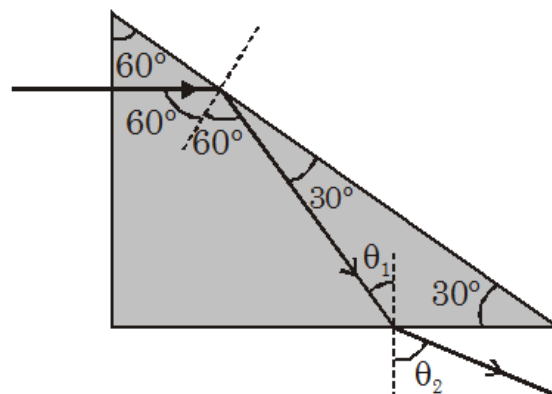
$$\therefore \theta_2 = \sin^{-1} \left(\frac{5}{8} \right)$$

for total internal reflection at P, we use

$$\frac{5}{3} \sin 60^\circ < n$$

$$n > \frac{5}{2\sqrt{3}}$$

6. **Ans (2.50)**



$$\mu_1 \sin \theta_1 = \mu_2 \sin \theta_2$$

$$\frac{5}{3} \sin 30^\circ = \frac{4}{3} \sin \theta_2$$

$$\therefore \theta_2 = \sin^{-1} \left(\frac{5}{8} \right)$$

for total internal reflection at P, we use

$$\frac{5}{3} \sin 60^\circ < n$$

$$n > \frac{5}{2\sqrt{3}}$$

PART-1 : PHYSICS

SECTION-II (ii)

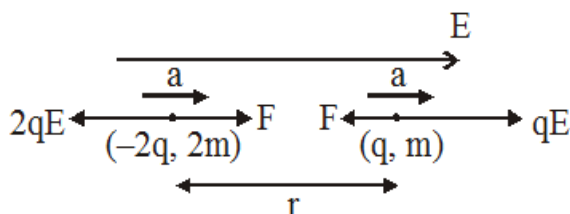
7. **Ans (35)**

Object must be at $2f$ distance of combination.

$$\frac{1}{f_{\text{net}}} = \frac{-2}{f_L} + \frac{1}{f_M}$$

8. **Ans (225)**

On releasing the particles, if their acceleration is same, they will stay at a constant distance
Acceleration of charged particle



$$a = \frac{F - 2qE}{2m} \text{ and } a = \frac{qE - F}{m}$$

$$\text{So, } F - 2qE = 2qE - 2F$$

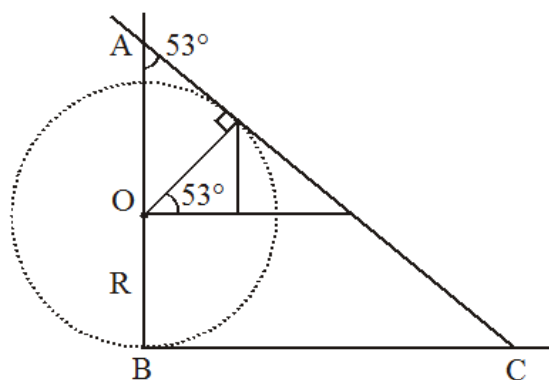
$$F = \frac{4qE}{3}$$

When they stay at a constant distance from each other

$$\frac{k \cdot q \cdot 2q}{r^2} = \frac{4qE}{3} \Rightarrow r = \sqrt{\frac{3kq}{2E}}$$

9. **Ans (60)**

$$\sin 53^\circ = \frac{R}{OA}$$



$$OA = \frac{5R}{4}$$

$$AB = \frac{9R}{4}$$

$$\tan 53 = \frac{BC}{AB}$$

$$\Rightarrow BC = \frac{9R}{4} \times \frac{4}{3}$$

$$\Rightarrow BC = 3R = 60 \text{ cm}$$

PART-2 : CHEMISTRY

SECTION-I (i)

2. **Ans (D)**

Solution. In cubic close packed structure of mixed oxides, the cations occupy holes while anions occupy primitives of the cube. In c.c.p, there is one octahedral and two tetrahedral voids per anion. So, for one oxygen atom, there is one octahedral and two tetrahedral holes.

(i) It is given that one half of the octahedral holes are occupied by B^{+x} ions. So :

Number of trivalent cations =

$$1 \times \frac{1}{2} = \frac{1}{2} \text{ (ii) It is given that one fifth of the tetrahedral voids are occupied by } A^{2+} \text{ ions. So :}$$

$$\text{Number of divalent cations} = 2 \times \frac{1}{5} = \frac{2}{5}$$

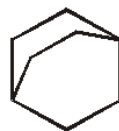
Hence, the formula of the compound = $A_{2/5}B_{1/2}O_1$
= $A_4B_5O_{10}$

$$\left[\because \frac{2}{5}, \frac{1}{2}, 1 = \frac{4 : 5 : 10}{10(\text{LCM})} = 4 : 5 : 10 \right]$$

PART-2 : CHEMISTRY

SECTION-I (ii)

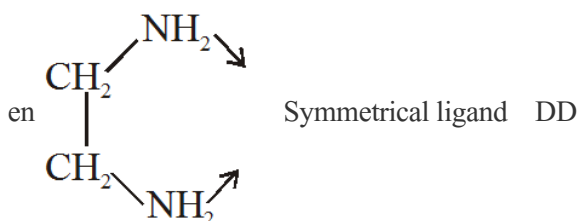
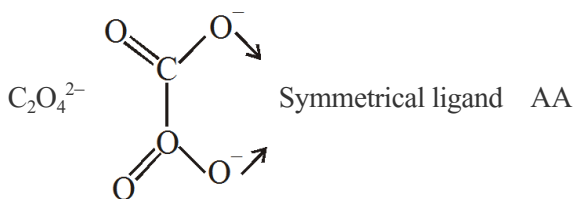
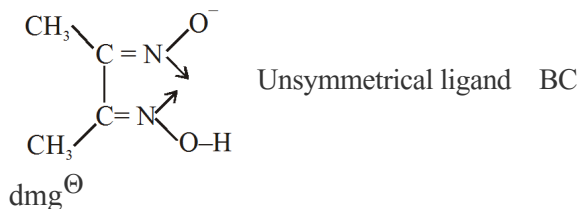
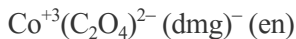
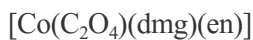
8. **Ans (A,B,C)**



PART-2 : CHEMISTRY

SECTION-II (i)

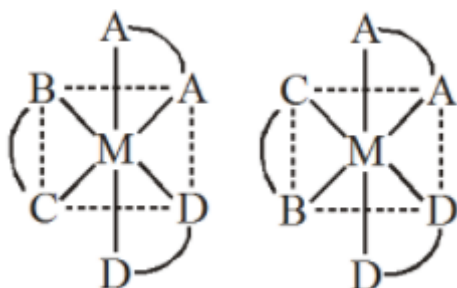
2. **Ans (2.00)**



Co-ordination compounds becomes



All three are bidentate ligands



2 Geometrical isomers are possible.

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (B)**

$$\lim_{h \rightarrow 0} \frac{(\log(2+h) - \log 2)3 \cdot (4^h - 1 - h)}{((8+h)^{1/3} - (4+3h)^{1/2}) \cdot \sin \pi(1+h)}$$

$$\lim_{h \rightarrow 0} \frac{3 \cdot \log\left(1 + \frac{h}{2}\right) \cdot (4^h - 1 - h)}{2 \left[\left(1 + \frac{h}{8}\right)^{1/3} - \left(1 + \frac{3h}{4}\right)^{1/2} \right] (-\sin \pi h)}$$

$$\lim_{h \rightarrow 0} \frac{3 \cdot \left(\log \frac{(1+\frac{h}{2})}{\frac{h}{2}} \cdot \frac{h}{2} \right) \cdot (4^h - 1 - h)}{2 \left[\left(1 + \frac{h}{24} + \dots\right) - \left(1 + \frac{3h}{8}\right) \right] \cdot \left(\frac{-\sin \pi h}{\pi h} \right) \pi h}$$

Using standard limit

$$\lim_{h \rightarrow 0} \frac{\left(\frac{3h}{2}\right) (4^h - 1 - h)}{2 \left(\frac{h}{3}\right) (\pi h)} = \lim_{h \rightarrow 0} \frac{9}{4\pi} \left(\frac{4^h - 1 - h}{h} \right)$$

Using LH rule

$$\lim_{h \rightarrow 0} \left(\frac{9}{4\pi} \right) \left(\frac{4^h \log 4 - 1}{1} \right)$$

Taking limit $\left(\frac{9}{4\pi} \right) (\log 4 - 1)$

2. **Ans (B)**

$$\frac{\sum \sin \theta_1}{\sin(\sum \theta_1)} = \frac{\sum \cos \theta_1}{\cos(\sum \theta_1)}$$

$$= \frac{\sin(\sum \theta_1) \cdot (\sum \sin \theta_1) + \cos(\sum \theta_1) (\sum \cos \theta_1)}{(\sin^2 \sum \theta_1) + (\cos^2 \sum \theta_1)} = 2$$

$$\Rightarrow \cos(\theta_1 + \theta_2) + \cos(\theta_2 + \theta_3) + \cos(\theta_3 + \theta_1) = 2$$

$$\Rightarrow \lambda = 2$$

$$\lim_{x \rightarrow 2} \frac{\sqrt{x^2 - 4}}{\sqrt{x} - 2 + \sqrt{x} - \sqrt{2}} = 2$$

3. **Ans (C)**

$$2f(\sin x) + \sqrt{2}f(\cos x) = \tan x \quad \dots(1)$$

$$2f(\cos x) + \sqrt{2}f(\sin x) = \cot x \quad \dots(2)$$

By solving

$$f(\cos x) = \cot x - \frac{\tan x}{\sqrt{2}}$$

$$f(t) = \frac{t}{\sqrt{1-t^2}} - \frac{\sqrt{1-t^2}}{\sqrt{2}t}$$

$$\sqrt{1-t} \times f(t) = \lim_{t \rightarrow 1} \frac{t}{\sqrt{1+t}} - \frac{\sqrt{1-t^2} \sqrt{(1-t)}}{\sqrt{2}t}$$

$$= \frac{1}{\sqrt{2}}$$

4. Ans (C)

$$4\cos x \cdot \cos 2x + 1 > 0$$

$$4\cos x (2\cos^2 x - 1) + 1 > 0$$

$$8\cos^3 x - 4\cos x + 1 > 0$$

$$8\cos^2 x \left(\cos x - \frac{1}{2} \right) + 4\cos x \left(\cos x - \frac{1}{2} \right) - 2 \left(\cos x - \frac{1}{2} \right) > 0$$

$$\text{Let } \cos x = t$$

$$\left(t - \frac{1}{2} \right) (8t^2 + 4t - 2) > 0$$

$$\left(t - \frac{1}{2} \right) (4t^2 + 2t - 1) > 0$$

$$t = \frac{-1 \pm \sqrt{5}}{4}$$

$$\therefore \cos x > \frac{1}{3}$$

$$\therefore 1 \geq \cos x > \frac{1}{3}$$

$$\therefore x \in \left(-\frac{\pi}{3}, \frac{\pi}{3} \right)$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (A,B)

$$\lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^{\frac{\tan x}{x - \tan x}} = \lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^{\frac{1}{\frac{x}{\tan x} - 1}} \text{ is}$$

$(\rightarrow 1) \rightarrow \infty$ format

$$\lim_{x \rightarrow 0} e^{\left(\frac{\tan x}{x} - 1 \right) \frac{\tan x}{x - \tan x}} = \lim_{x \rightarrow 0} e^{-\frac{\tan x}{x}} = e^{-1} = \frac{1}{e}$$

$$\text{Now, } \lim_{x \rightarrow \infty} \left(\frac{2x^2 + 1}{4x^2 - 1} \right)^{\frac{x^2}{2+x}} = \left(\rightarrow \frac{1}{2} \right)^{\rightarrow \infty} = 0$$

$$f_1(x) = \frac{x}{3} + 2 \quad \forall x \in \mathbb{R}; f_n(x) = f_1(f_{n-1}(x))$$

$$\lim_{n \rightarrow \infty} f_n(x) = \lim_{n \rightarrow \infty} f_1(f_{n-1}(x))$$

$$\lim_{n \rightarrow \infty} f_n(x) = \left(\lim_{n \rightarrow \infty} f_{n-1}(x) \right) \quad [\because f_1(x) \text{ is continuous}]$$

$$k = f_1(k)$$

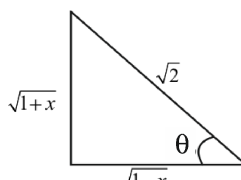
$$k = \frac{k}{3} + 2 \Rightarrow k = 3$$

6. Ans (A,B)

$$\sin^{-1} \sqrt{\frac{1+x}{2}} - \sqrt{2-x} = \cot^{-1}(\tan \sqrt{2-x}) - \sin^{-1} \sqrt{\frac{1-x}{2}}$$

$$\begin{matrix} \Downarrow & \Downarrow & \Downarrow & \Downarrow \\ -1 \leq x \leq 1 & x \leq 2 & x \leq 2 & -1 \leq x \leq 1 \end{matrix}$$

Domain : $x \in [-1, 1]$



$$\sin^{-1} \sqrt{\frac{1+x}{2}} + \sin^{-1} \sqrt{\frac{1-x}{2}} - \cot^{-1}(\tan \sqrt{2-x}) = \sqrt{2-x}$$

$$\frac{\pi}{2} - \cot^{-1}(\tan \sqrt{2-x}) = \sqrt{2-x}$$

$$\tan^{-1}(\tan \sqrt{2-x}) = \sqrt{2-x}$$

$$\text{Let } \sqrt{2-x} = \alpha$$

$$\text{where } \alpha \in [1, \sqrt{3}] - \frac{\pi}{2}; \text{ as } x \in [-1, 1]$$

$$\Rightarrow \tan^{-1}(\tan \alpha) = \alpha \quad \dots (1)$$

$$\text{Case I : } \alpha \in \left[1, \frac{\pi}{2} \right)$$

$$\tan^{-1}(\tan \alpha) = \alpha \text{ happens } \forall \alpha \in \left[1, \frac{\pi}{2} \right)$$

$$1 \leq \sqrt{2-x} < \frac{\pi}{2} \text{ or } 2 - \frac{\pi^2}{4} < x \leq 1$$

$$\text{Case II : } \alpha \in \left[\frac{\pi}{2}, \sqrt{3} \right)$$

$$\tan^{-1}(\tan \alpha) \text{ becomes } \alpha - \pi$$

$$\text{Hence (1) becomes } \alpha - \pi = \alpha \text{ (Not possible)}$$

7. Ans (A,B,D)

$f(x)$ must be surjective

$h(x)$ must be injective

8. Ans (A,B,C,D)

$$y \sin \theta = x$$

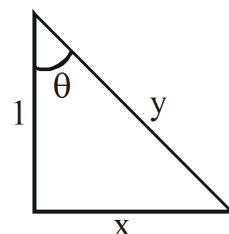
$$y \cos \theta = 1$$

$$(y^n - x^n) = \frac{1 - \sin^n \theta}{\cos^n \theta}$$

$$\lim_{\theta \rightarrow \frac{\pi}{2}} \frac{1 - \sin \theta}{\cos \theta} = 0,$$

$$\lim_{\theta \rightarrow \frac{\pi}{2}} \frac{1 - \sin^2 \theta}{\cos^2 \theta} = 1$$

$$\lim_{\theta \rightarrow \frac{\pi}{2}} \frac{1 - \sin^3 \theta}{\cos^3 \theta} = \infty \Rightarrow \text{Limit does not exist.}$$



9. Ans (A,B,C,D)

All are incorrect statements

10. Ans (B,D)

$$T_r = \cot^{-1} \left(r^2 + \sin^2 \frac{2r\pi}{3} \right) = \cot^{-1} \left(r^2 + \frac{3}{4} \right)$$

$$\left[\because \sin^2 \frac{2r\pi}{3} = \frac{3}{4} \right]$$

$$= \tan^{-1} \left(\frac{1}{r^2 + \frac{3}{4}} \right) = \tan^{-1} \left(\frac{1}{1 + r^2 - \frac{1}{4}} \right)$$

$$= \tan^{-1} \left(\frac{(r + \frac{1}{2}) - (r - \frac{1}{2})}{1 + (r + \frac{1}{2})(r - \frac{1}{2})} \right)$$

$$T_r = \tan^{-1} \left(r + \frac{1}{2} \right) - \tan^{-1} \left(r - \frac{1}{2} \right)$$

$$S_n = \sum_{r=1}^n T_r = \tan^{-1} \left(n + \frac{1}{2} \right) - \tan^{-1} \left(\frac{1}{2} \right)$$

$$\therefore S_\infty = \frac{\pi}{2} - \tan^{-1} \left(\frac{1}{2} \right)$$

$$= \cot^{-1} \left(\frac{1}{2} \right) = \tan^{-1} (2) = \theta$$

Also

$$\cot \left(\theta - \frac{\pi}{4} \right) = \frac{\cot \theta \cdot \cot \frac{\pi}{4} + 1}{\cot \frac{\pi}{4} - \cot \theta} = \frac{\frac{1}{2} + 1}{1 - \frac{1}{2}} = 3$$

$$\Rightarrow \theta = \frac{\pi}{4} + \cot^{-1} (3) = \cot^{-1} (1) + \cot^{-1} (3)$$

$$\frac{1}{2} \sin^{-1} \left(\frac{4}{5} \right) < \frac{\pi}{4} \text{ but } \tan^{-1} 2 > \frac{\pi}{4}$$

PART-3 : MATHEMATICS

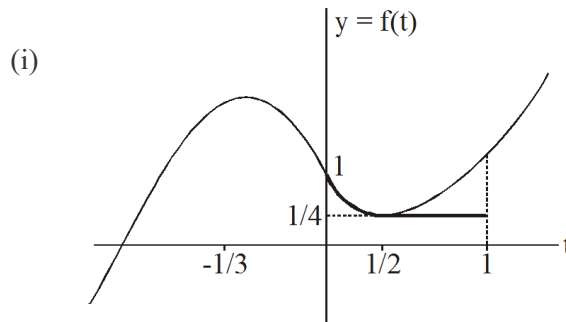
SECTION-II (i)

1. Ans (0.50)

$$f(x) = 4x^3 - x^2 - 2x + 1 ;$$

$$g(x) = \begin{cases} \min \{f(t) ; 0 \leq t \leq x\} & ; 0 \leq x \leq 1 \\ 3-x & ; 1 < x \leq 2 \end{cases}$$

$$g(x) = \begin{cases} 4x^3 - x^2 - 2x + 1 & ; 0 \leq x \leq \frac{1}{2} \\ \frac{1}{4} & ; \frac{1}{2} < x \leq 1 \\ 3-x & ; 1 < x \leq 2 \end{cases}$$



$$g\left(\frac{1}{4}\right) = \frac{1}{2}$$

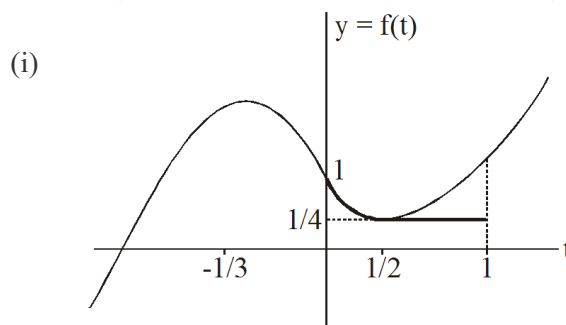
$$(ii) g\left(\frac{3}{4}\right) = \frac{1}{4}$$

2. Ans (0.25)

$$f(x) = 4x^3 - x^2 - 2x + 1 ;$$

$$g(x) = \begin{cases} \min \{f(t) ; 0 \leq t \leq x\} & ; 0 \leq x \leq 1 \\ 3-x & ; 1 < x \leq 2 \end{cases}$$

$$g(x) = \begin{cases} 4x^3 - x^2 - 2x + 1 & ; 0 \leq x \leq \frac{1}{2} \\ \frac{1}{4} & ; \frac{1}{2} < x \leq 1 \\ 3-x & ; 1 < x \leq 2 \end{cases}$$



$$g\left(\frac{1}{4}\right) = \frac{1}{2}$$

$$(ii) g\left(\frac{3}{4}\right) = \frac{1}{4}$$

3. **Ans (0.00)**

$$f(x) = (2x + 1)^3$$

4. **Ans (1.00)**

$$f(x) = (2x + 1)^3$$

5. **Ans (6.00)**

$$\text{Given } a_1 = 1$$

$$\Rightarrow a_2 = \sin a_1 = \sin 1$$

$$a_3 = \sin a_2 = \sin(\sin 1)$$

$\vdots$

$$a_5 = \sin a_4 = \sin(\sin(\sin(\sin 1)))$$

$$\Rightarrow a_1, a_2, a_3, \dots, a_n \text{ is a decreasing sequence}$$

$$\text{hence as } n \rightarrow \infty \lim_{n \rightarrow \infty} a_n = 0$$

$$\text{Let } a_n = y$$

$$\therefore \text{ as } n \rightarrow \infty ; y \rightarrow 0$$

$$\lim_{y \rightarrow 0} \frac{2^{2y} - 2(2^y)(3^y) + 2^{2y}}{\cos y + 1 - e^y - e^{-y}}$$

$$-\lim_{y \rightarrow 0} \frac{(2^y - 3^y)^2}{(e^y - e^{-y} - 2) + (1 - \cos y)} = \frac{-2}{3} \ln^2 \left(\frac{2}{3} \right)$$

6. **Ans (4.00)**

$$\text{Given } a_1 = 1$$

$$\Rightarrow a_2 = \sin a_1 = \sin 1$$

$$a_3 = \sin a_2 = \sin(\sin 1)$$

$\vdots$

$$a_5 = \sin a_4 = \sin(\sin(\sin(\sin 1)))$$

$$\Rightarrow a_1, a_2, a_3, \dots, a_n \text{ is a decreasing sequence}$$

$$\text{hence as } n \rightarrow \infty \lim_{n \rightarrow \infty} a_n = 0$$

$$\text{Let } a_n = y$$

$$\therefore \text{ as } n \rightarrow \infty ; y \rightarrow 0$$

$$\lim_{y \rightarrow 0} \frac{2^{2y} - 2(2^y)(3^y) + 2^{2y}}{\cos y + 1 - e^y - e^{-y}}$$

$$-\lim_{y \rightarrow 0} \frac{(2^y - 3^y)^2}{(e^y - e^{-y} - 2) + (1 - \cos y)} = \frac{-2}{3} \ln^2 \left(\frac{2}{3} \right)$$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. **Ans (1)**

$$a_1 = 1, a_n = n(1 + a_{n-1})$$

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{a_1} \right) \left(1 + \frac{1}{a_2} \right) \dots \left(1 + \frac{1}{a_n} \right)$$

$$\lim_{n \rightarrow \infty} \frac{(1 + a_1)(1 + a_2) \dots (1 + a_n)}{a_1 a_2 a_3 \dots a_n}$$

$$\ell = \lim_{n \rightarrow \infty} \frac{\left(\frac{a_2}{2} \right) \left(\frac{a_3}{3} \right) \dots \left(\frac{a_{n+1}}{n+1} \right)}{a_1 a_2 a_3 \dots a_n} = \frac{a_{n+1}}{(n+1)!}$$

$$\ell = \lim_{n \rightarrow \infty} \frac{(n+1)(1 + a_n)}{(n+1)!} = \frac{1}{n!} + \frac{a_n}{n!}$$

$$\ell = \lim_{n \rightarrow \infty} \frac{1}{n!} + \frac{n(1 + a_{n-1})}{n!}$$

$$\ell = \lim_{n \rightarrow \infty} \frac{1}{n!} + \frac{1}{(n-1)!} + \frac{a_{n-1}}{(n-1)!}$$

Continuing in same way;

$$\ell = \lim_{n \rightarrow \infty} \frac{1}{n!} + \frac{1}{(n-1)!} + \frac{1}{(n-2)!} + \dots + \frac{1}{1!} + \frac{a_1}{1!}$$

$$\ell = \lim_{n \rightarrow \infty} \frac{1}{n!} + \frac{1}{1!} + \frac{1}{2!} + \dots + \frac{1}{n!} \quad (\text{In reverse order})$$

As $n \rightarrow \infty$, series on RHS converges to e

8. **Ans (4)**

$$\sin^{-1} x + \sin^{-1} y \geq (x+y)^2 - 2[xy + x - 1 + y] + \pi$$

$$\sin^{-1} x + \sin^{-1} y \geq x^2 + y^2 - 2x - 2y + 1 + 1 + \pi$$

$$\left(\frac{\pi}{2} - \sin^{-1} x \right) + \left(\frac{\pi}{2} - \sin^{-1} y \right) + (x-1)^2 + (y-1)^2 \leq 0$$

....(i)

All terms on LHS are ≥ 0 , so (i) is

$$\text{valid when } \sin^{-1} x = \sin^{-1} y = \frac{\pi}{2}; x = y = 1$$

9. **Ans (3)**

$$\lim_{x \rightarrow 0} \frac{a + \left(x - \frac{x^2}{2} \right) + bx}{x^2} = \ell$$

$$\lim_{x \rightarrow 0} \frac{a + (b+1)x - \frac{x^2}{2}}{x^2} = \ell$$

$$\text{If limit exist then, } a = 0, b = -1, \ell = -\frac{1}{2}$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

REVIEW TEST

15-03-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TNAS]

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,B,C	B,C,D	A,C	A,C,D	A,C	A,C,D
SECTION-I (ii)	Q.	7	8	9	10		
	A.	D	D	A	B		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	1.50	0.66 to 0.67	1.00	1.24	1.73	41.00
SECTION-II (ii)	Q.	7	8	9			
	A.	3	12	9			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	B,C,D	A,C,D	A,C,D	C,D	B,D	A,B,C
SECTION-I (ii)	Q.	7	8	9	10		
	A.	C	D	B	A		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	3.00	8.00	1.00	3.00	30.00	40.00
SECTION-II (ii)	Q.	7	8	9			
	A.	7	36	400			

PART-3 : MATHEMATICS

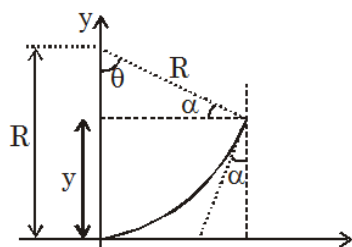
SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	C	C	A,B	A,C,D	A,B,C,D	A,B,C
SECTION-I (ii)	Q.	7	8	9	10		
	A.	D	C	D	B		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	3.00	0.00	0.75	2.00	7.00	2.00
SECTION-II (ii)	Q.	7	8	9			
	A.	561	8	429			

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A,B,C)



$$\mu_0 \sin 90^\circ = \mu \sin \alpha$$

$$\frac{\mu_0}{\sin \alpha} = \mu$$

$$\mu = \frac{\mu_0 R}{R - y}$$

$$\mu_0 = 1$$

$$\sin \alpha = \frac{R - y}{R}$$

$$y = \frac{R}{2}$$

$$\sin \alpha = \frac{1}{2}$$

$$\alpha = 30^\circ$$

$$\text{unit vector is } \sin \alpha \hat{i} + \cos \alpha \hat{j}$$

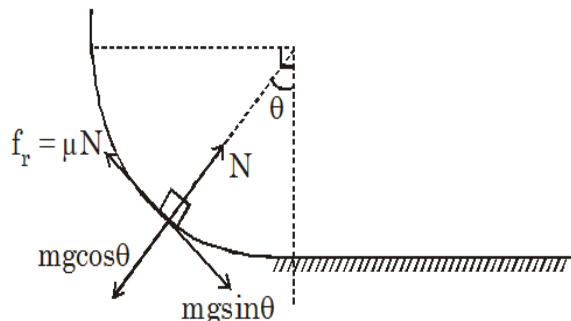
$$\frac{5}{2} = \mu = \frac{R}{R - y} \Rightarrow 5(R - y) = 2R$$

$$\Rightarrow y = \frac{3R}{5} = \cos \theta_{\max} = \frac{2}{5}$$

2. Ans (B,C,D)

$$N - mg \cos \theta = \frac{mv^2}{r}$$

$$N = m \left(g \cos \theta + \frac{v^2}{r} \right)$$



as speed is constant

$$mg \sin \theta = \mu m \left(g \cos \theta + \frac{v^2}{r} \right)$$

$$\mu = \frac{\sin \theta}{\cos \theta + \frac{v^2}{gR}}$$

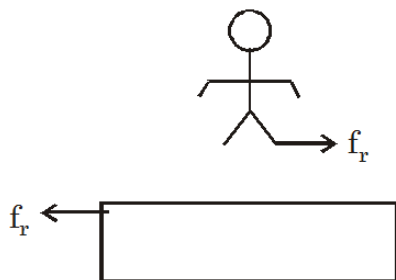
$$\text{Power due to friction} = \vec{F} \cdot \vec{v} = -mg \sin \theta v$$

Applying work energy theorem on block from A to C

$$k_f - k_i = \omega_g + \omega_{fr}$$

$$0 = +mgR = \omega_{fr} \Rightarrow \omega_{fr} = -mgR$$

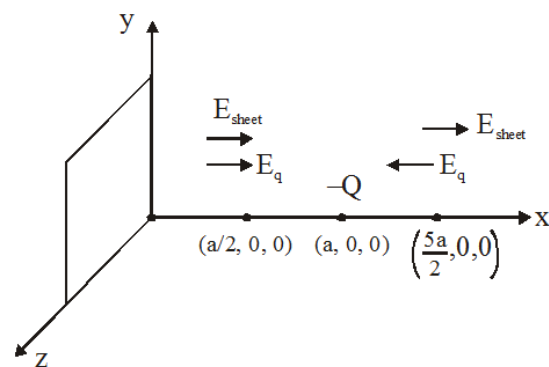
3. Ans (A,C)



Work done by friction on plank is positive because displacement and direction of friction is parallel.

Work done by friction on man is negative because friction is acting towards right but point of application of force is moving towards left.

4. Ans (A,C,D)



From graph electric field at $(2a, 0, 0)$ is zero

$$\text{So } \frac{\sigma}{2\epsilon_0} = \frac{KQ}{a^2} \Rightarrow Q = \sigma^2 \pi a^2$$

Now electric field at

$$\left(\frac{a}{2}, 0, 0 \right) = \frac{\sigma}{2\epsilon_0} + \frac{K\sigma 2\pi a^2}{(a/2)^2}$$

$$= \frac{5\sigma}{2\epsilon_0}$$

$$\text{Net field at } \left(\frac{5a}{2}, 0, 0 \right) = \frac{\sigma}{2\epsilon_0} - \frac{K\sigma 2\pi a^2}{(3a/2)^2}$$

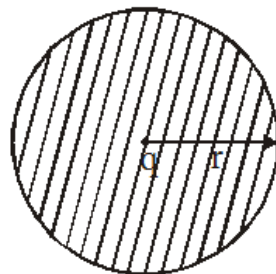
5. Ans (A,C)

Charge on sphere of radius as

$$\begin{aligned} &= \int_0^\infty 4\pi r^2 dr (3) \\ &= \frac{-q}{4\pi a} \cdot \frac{e^{-r/a}}{r^2} \times 4\pi r^2 dr \\ &= \frac{q}{a} \int_0^\infty e^{-\frac{r}{a}} dr = -q \end{aligned}$$

so net charge in space = $q - q = 0$

Electric field at point P is



$$E = \frac{kq_{in}}{r^2} \text{ due to spherical symmetry}$$

$$q_{in} = q + (e^{-\frac{r}{a}} - 1)q$$

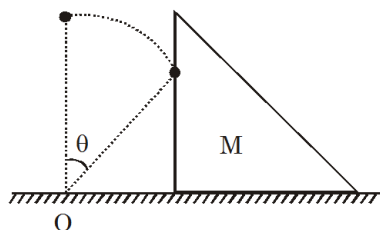
$$q_{in} = qe^{-\frac{r}{a}}$$

$$E = \frac{qe^{-\frac{r}{a}}}{4\pi\epsilon_0 r^2}$$

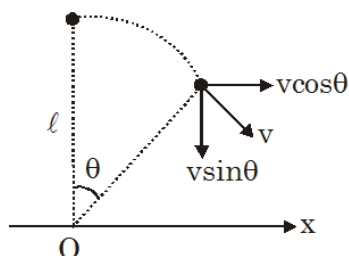
PART-1 : PHYSICS

SECTION-I (ii)

7. Ans (D)



∴ Condition when contact is lost is
 a_x = horizontal acceleration of $m = 0$
 Velocity of M will be equal to horizontal component of velocity of m .



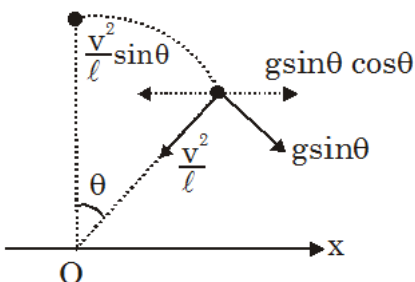
∴ Velocity of M is $v_x = v \cos \theta$
 ∴ loss in P.E of m = gain in K.E of m + gain in K.E of M

$$mg\ell(1 - \cos \theta) = \frac{1}{2}mv^2 + \frac{1}{2}M(v \cos \theta)^2$$

$$\Rightarrow v = \frac{2mg\ell(1 - \cos \theta)}{m + M \cos^2 \theta} \quad \dots(i)$$

The acceleration of mass m in horizontal direction is

$$a_x = g \sin \theta \cdot \cos \theta - \frac{v^2}{\ell} \sin \theta \quad (\text{see figure})$$



For losing contact $a_x = 0$

$$\therefore g \sin \theta \cdot \cos \theta - \frac{2mg(1 - \cos \theta) \sin \theta}{m + M \cos^2 \theta}$$

[using (i)]

$$\Rightarrow 2mg - 2mg \cos \theta - mg \cos \theta - Mg \cos^3 \theta = 0 \quad [\sin \theta = 0 \text{ is not acceptable}]$$

It is given that this happens when $\theta = \frac{\pi}{3}$

$$\Rightarrow \frac{1}{2}m = \frac{M}{8}$$

$$\Rightarrow M = 4m$$

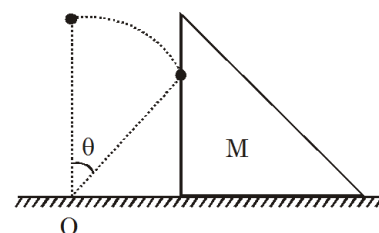
(ii) and (i)

$$v^2 = \frac{2mg\ell(1 - \cos \pi/3)}{m + 4m(\cos \pi/3)^2} = \frac{g\ell}{2}$$

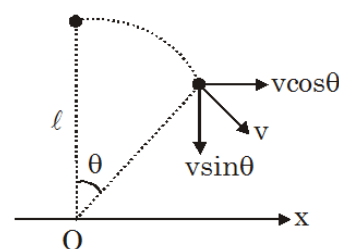
∴ Speed of M at the instant is

$$v_x = v \cos \theta = \sqrt{\frac{g\ell}{2}} \cos \frac{\pi}{3} = \frac{1}{2} \sqrt{\frac{g\ell}{2}}$$

8. Ans (D)



∴ Condition when contact is lost is
 a_x = horizontal acceleration of $m = 0$
 Velocity of M will be equal to horizontal component of velocity of m .



∴ Velocity of M is $v_x = v \cos \theta$

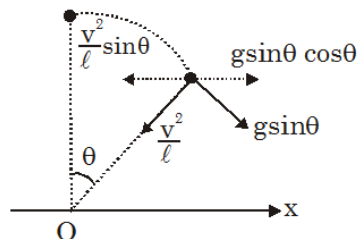
∴ loss in P.E of m = gain in K.E of m + gain in K.E of M

$$mg\ell(1 - \cos \theta) = \frac{1}{2}mv^2 + \frac{1}{2}M(v \cos \theta)^2$$

$$\Rightarrow v = \frac{2mg\ell(1 - \cos \theta)}{m + M \cos^2 \theta} \quad \dots(i)$$

The acceleration of mass m in horizontal direction is

$$a_x = g \sin \theta \cdot \cos \theta - \frac{v^2}{\ell} \sin \theta \quad (\text{see figure})$$



For losing contact $a_x = 0$

$$\therefore g \sin \theta \cdot \cos \theta - \frac{2mg(1 - \cos \theta) \sin \theta}{m + M \cos^2 \theta}$$

[using (i)]

$$\Rightarrow 2mg - 2mg\cos\theta - mg\cos\theta - Mg\cos^3\theta$$

$$= 0 \text{ [}\sin\theta = 0 \text{ is not acceptable]}$$

It is given that this happens when $\theta = \frac{\pi}{3}$

$$\Rightarrow \frac{1}{2}m = \frac{M}{8}$$

$$\Rightarrow M = 4m$$

(ii) and (i)

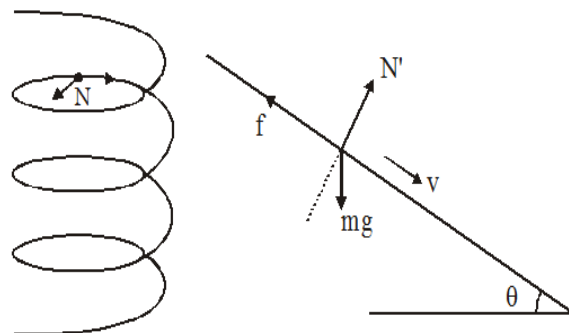
$$v^2 = \frac{2mg\ell(1 - \cos\pi/3)}{m + 4m(\cos\pi/3)^2} = \frac{g\ell}{2}$$

$\therefore$ Speed of M at the instant is

$$v_x = v \cos\theta = \sqrt{\frac{g\ell}{2}} \cos\frac{\pi}{3} = \frac{1}{2}\sqrt{\frac{g\ell}{2}}$$

9. Ans (A)

The helix wire is inclined to the horizontal at an angle given by



$$\tan\theta = \frac{\rho}{2\pi R} \quad \dots(i)$$

The normal force applied by the wire on the bead can be through to have two components.

N = directed horizontally towards axis of helix. (In the second figure given above, this force is normal to the plane the drawing pointing towards you)

N' = directed perpendicular to the velocity in the plane of above figure.

Force equations are (when the bead acquires constant speed)

$$f = mg \sin\theta \quad \dots(ii)$$

$$N' = mg \cos\theta \quad \dots(iii)$$

$$N = \frac{mv_H^2}{R}$$

$$\Rightarrow N = \frac{mv^2 \cos^2\theta}{R} \quad \dots(iv)$$

Where v_H = Horizontal component of velocity
= $v \cos\theta$

The resultant of two components of normal forces is

$$N_0 = \sqrt{N^2 + N'^2}$$

$$= \sqrt{(mg \cos\theta)^2 + \left(\frac{mv^2 \cos^2\theta}{R}\right)^2}$$

Now friction $f = \mu N$

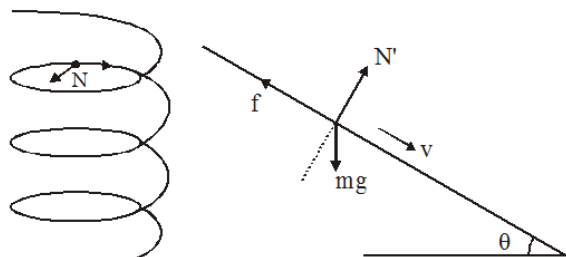
$\therefore$ from (ii)

$$\mu mg \cos\theta \sqrt{1 + \frac{v^4}{R^2 g^2} \cos^2\theta} = mg \sin\theta$$

$$\therefore \mu = \frac{\tan\theta}{\sqrt{1 + \left(\frac{v^2 \cos\theta}{Rg}\right)^2}}$$

10. Ans (B)

The helix wire is inclined to the horizontal at an angle given by



$$\tan \theta = \frac{\rho}{2\pi R} \quad \dots(i)$$

The normal force applied by the wire on the bead can be through to have two components. N = directed horizontally towards axis of helix. (In the second figure given above, this force is normal to the plane the drawing pointing towards you)

N' = directed perpendicular to the velocity in the plane of above figure.

Force equations are (when the bead acquires constant speed)

$$f = mg \sin \theta \quad \dots(ii)$$

$$N' = mg \cos \theta \quad \dots(iii)$$

$$N = \frac{mv_H^2}{R}$$

$$\Rightarrow N = \frac{mv^2 \cos^2 \theta}{R} \quad \dots(iv)$$

Where v_H = Horizontal component of velocity
= $v \cos \theta$

The resultant of two components of normal forces is

$$N_0 = \sqrt{N^2 + N'^2}$$

$$= \sqrt{(mg \cos \theta)^2 + \left(\frac{mv^2 \cos^2 \theta}{R}\right)^2}$$

Now friction $f = \mu N$

$\therefore$ from (ii)

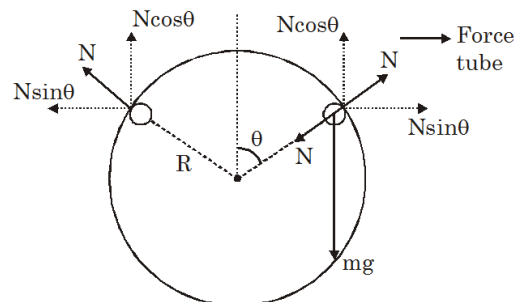
$$\mu mg \cos \theta \sqrt{1 + \frac{v^4}{R^2 g^2} \cos^2 \theta} = mg \sin \theta$$

$$\therefore \mu = \frac{\tan \theta}{\sqrt{1 + \left(\frac{v^2 \cos \theta}{Rg}\right)^2}}$$

PART-1 : PHYSICS

SECTION-II (i)

1. Ans (1.50)



When $2N \cos \theta > Mg$ (weight of tube), the tube will begin to rise.

Considering the dynamics of ball

$$\frac{mv^2}{R} = mg \cos \theta + N \quad \dots(i)$$

From conservation of energy

$$\frac{mv^2}{R} = mgR(1 - \cos \theta) \quad \dots(ii) \text{ [For each ball]}$$

From (i) and (ii), eliminating v^2 , we get

$$N = mg(2 - 3 \cos \theta)$$

For tube to rise

$$2N \cos \theta \geq Mg$$

$$2m \cos \theta (2 - 3 \cos \theta) \geq M \quad \dots(iii)$$

For the tube to just start rising, the maximum value of

$Z = \cos \theta (2 - 3 \cos \theta)$ must satisfy the above inequality.

Z is maximum if

$$\frac{dz}{d\theta} = -2 \sin \theta + 3 \times 2 \cos \theta \cdot \sin \theta = 0$$

$$\Rightarrow \cos \theta = \frac{1}{3}$$

$$\therefore Z_{\max} = \frac{1}{3} \times \left(2 - 3 \times \frac{1}{3}\right) = \frac{1}{3}$$

$$\therefore \text{from (iii)} \quad 2m \frac{1}{3} \geq M$$

$$\Rightarrow m \geq \frac{3M}{2} \text{ for the tube of rise.}$$

(ii) Equation (iii) gives the value of θ .

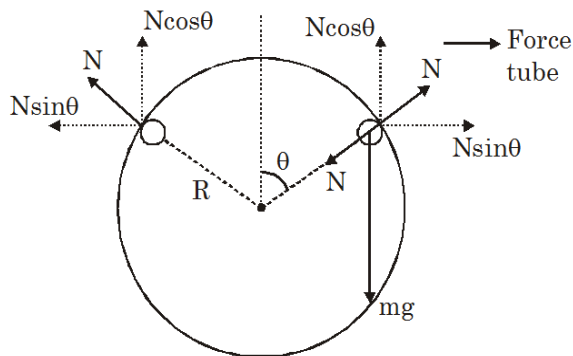
When $M = 0$,

$$2m \cos \theta (2 - 3 \cos \theta) = 0 \quad \text{[In limiting case]}$$

$$\Rightarrow \cos \theta = \frac{2}{3}$$

$$\theta = \cos^{-1} \left(\frac{2}{3}\right)$$

2. Ans (0.66 to 0.67)



When $2N \cos \theta > Mg$ (weight of tube), the tube will begin to rise.

Considering the dynamics of ball

$$\frac{mv^2}{R} = mg \cos \theta + N \quad \dots(i)$$

From conservation of energy

$$\frac{mv^2}{R} = mgR(1 - \cos \theta) \quad \dots(ii) \text{ [For each ball]}$$

From (i) and (ii), eliminating v^2 , we get

$$N = mg(2 - 3 \cos \theta)$$

For tube to rise

$$2N \cos \theta \geq Mg$$

$$2m \cos \theta (2 - 3 \cos \theta) \geq M \quad \dots(iii)$$

For the tube to just start rising, the maximum value of

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Z is maximum if

$$\frac{dz}{d\theta} = -2 \sin \theta + 3 \times 2 \cos \theta \cdot \sin \theta = 0$$

$$\Rightarrow \cos \theta = \frac{1}{3}$$

$$\therefore Z_{\max} = \frac{1}{3} \times \left(2 - 3 \times \frac{1}{3} \right) = \frac{1}{3}$$

$$\therefore \text{from (iii)} \quad 2m \frac{1}{3} \geq M$$

$$\Rightarrow m \geq \frac{3M}{2} \text{ for the tube of rise.}$$

(ii) Equation (iii) gives the value of θ .

When $M = 0$,

$$2m \cos \theta (2 - 3 \cos \theta) = 0 \quad \text{[In limiting case]}$$

$$\Rightarrow \cos \theta = \frac{2}{3}$$

$$\theta = \cos^{-1} \left(\frac{2}{3} \right)$$

3. Ans (1.00)

Let v = speed at B.

When rod becomes horizontal, we can write its

Electrostatic potential energy as

$$U_f = \int_0^L V_x \lambda dx \quad [V_x = \text{potential at distance } x \text{ from}$$

$$\text{end A} = V_0 - E_0 x]$$

$$= \int_0^L \lambda (V_0 - E_0 x) dx = \lambda V_0 L - \lambda E_0 \frac{L^2}{2}$$

Loss in electrostatic potential energy

$$U_i - U_f = \lambda E_0 \frac{L^2}{2}$$

This is equal to gain in gravitational potential energy.

$$\lambda E_0 \frac{L^2}{2} = Mg \frac{L}{2} \Rightarrow E_0 = \frac{Mg}{\lambda L}$$

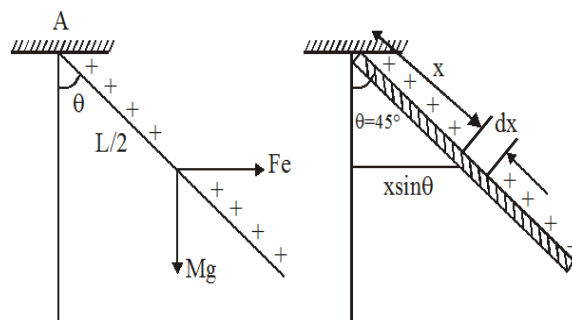
The electric force on the rod is $F_e = \lambda L E_0$. Force can be considered at centre for writing the torque. In equilibrium,

$$\tau_{Fe} = \tau_{mg} \text{ [torque about A]}$$

$$\lambda L E_0 \frac{L}{2} \cos \theta = Mg \frac{L}{2} \sin \theta$$

$$\text{Putting } E_0 = \frac{Mg}{\lambda L} \text{ we get } \theta = 45^\circ$$

The rod has maximum KE at this position.



KE = (loss in electrostatic PE) - (gain in gravitational PE)

$$= \int_0^L E_0 x \sin \theta \lambda dx - Mg \frac{L}{2} (1 - \cos \theta)$$

$$= \frac{MgL}{2\sqrt{2}} - Mg \frac{L}{2} \left(1 - \frac{1}{\sqrt{2}} \right)$$

$$\therefore \frac{1}{2} \left(\frac{ML^2}{3} \right) \omega^2 = MgL \left(\frac{\sqrt{2} - 1}{2} \right)$$

$$\Rightarrow \omega = \sqrt{3(\sqrt{2} - 1) \frac{g}{L}}$$

4. Ans (1.24)

Let v = speed at B.

When rod becomes horizontal, we can write its Electrostatic potential energy as

$$U_f = \int_0^L V_x \lambda dx \quad [V_x = \text{potential at distance } x \text{ from end A} = V_0 - E_0 x]$$

from end A = $V_0 - E_0 x$

$$= \int_0^L \lambda(V_0 - E_0 x) dx = \lambda V_0 L - \lambda E_0 \frac{L^2}{2}$$

Loss in electrostatic potential energy

$$U_i - U_f = \lambda E_0 \frac{L^2}{2}$$

This is equal to gain in gravitational potential energy.

$$\lambda E_0 \frac{L^2}{2} = Mg \frac{L}{2} \Rightarrow E_0 = \frac{Mg}{\lambda L}$$

The electric force on the rod is $F_e = \lambda L E_0$.

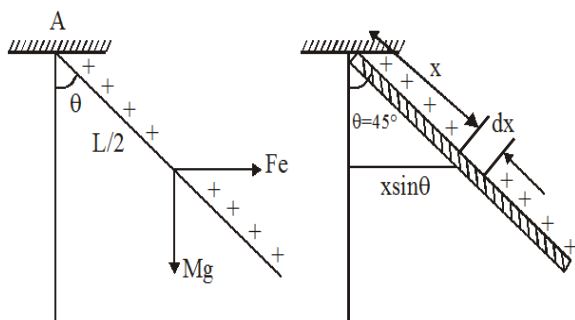
Force can be considered at centre for writing the torque. In equilibrium,

$$\tau_{Fe} = \tau_{mg} \quad [\text{torque about A}]$$

$$\lambda L E_0 \frac{L}{2} \cos \theta = Mg \frac{L}{2} \sin \theta$$

$$\text{Putting } E_0 = \frac{Mg}{\lambda L} \text{ we get } \theta = 45^\circ$$

The rod has maximum KE at this position.



KE = (loss in electrostatic PE) - (gain in gravitational PE)

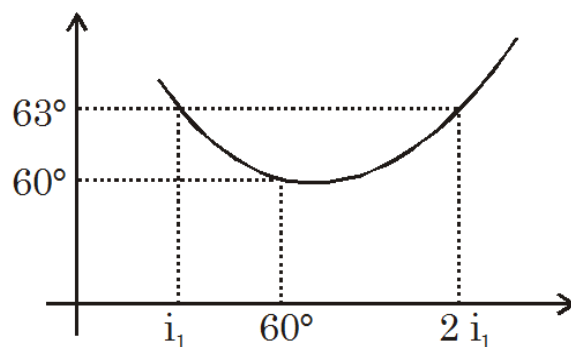
$$= \int_0^L E_0 x \sin \theta \lambda dx - Mg \frac{L}{2} (1 - \cos \theta)$$

$$= \frac{MgL}{2\sqrt{2}} - Mg \frac{L}{2} \left(1 - \frac{1}{\sqrt{2}}\right)$$

$$\therefore \frac{1}{2} \left(\frac{ML^2}{3} \right) \omega^2 = MgL \left(\frac{\sqrt{2}-1}{2} \right)$$

$$\Rightarrow \omega = \sqrt{3(\sqrt{2}-1) \frac{g}{L}}$$

5. Ans (1.73)



The minimum deviation

$$60^\circ = 2 \times 60^\circ - A$$

$$A = 60^\circ$$

Now for the deviation

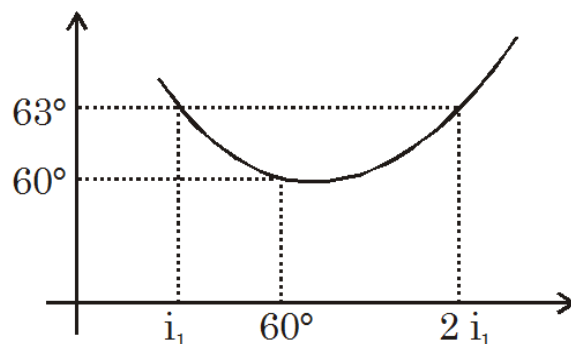
$$\delta = 63^\circ$$

$$\delta = i + e - A$$

$$63^\circ = i_1 + 2i_1 - 60^\circ$$

$$3i_1 = 123^\circ \Rightarrow i_1 = 41^\circ$$

6. Ans (41.00)



The minimum deviation

$$60^\circ = 2 \times 60^\circ - A$$

$$A = 60^\circ$$

Now for the deviation

$$\delta = 63^\circ$$

$$\delta = i + e - A$$

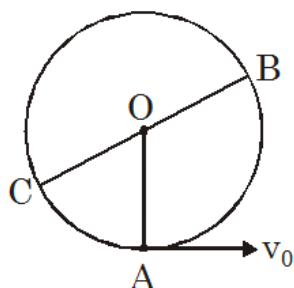
$$63^\circ = i_1 + 2i_1 - 60^\circ$$

$$3i_1 = 123^\circ \Rightarrow i_1 = 41^\circ$$

PART-1 : PHYSICS

SECTION-II (ii)

7. Ans (3)



Let v = speed at B.

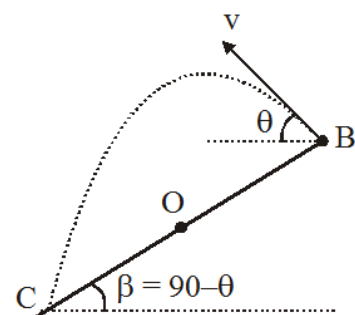
Tension in the string at B = 0

$$\therefore mg \cos \theta = \frac{mv^2}{l}$$

$$\Rightarrow v^2 = gl \cos \theta \quad \dots(i)$$

After B, the motion of particle is that of a projectile. One can consider it a projectile thrown down the incline B.

The particle hits the incline at C.



$$\frac{2gl \cos \theta \cdot \sin 90^\circ \cos \theta}{g \sin^2 \theta} = 2l \cot^2 \theta$$

$$\therefore 2l = 2l \cot^2 \theta$$

$$\therefore \cot^2 \theta = 1$$

$$\text{Or, } \theta = 45^\circ$$

Now we can use conservation of energy between points A and B, which gives

$$\frac{1}{2}mv_0^2 = \frac{1}{2}mv^2 + mg\ell(1 + \cos \theta)$$

$$\Rightarrow v_0^2 = v^2 + 2g\ell(1 + \cos \theta)$$

Using (i)

$$v_0^2 = 3g\ell \cos \theta + 2g\ell$$

$$= g\ell \left[3 \times \frac{1}{\sqrt{2}} + 2 \right] \quad [\because \theta = 45^\circ]$$

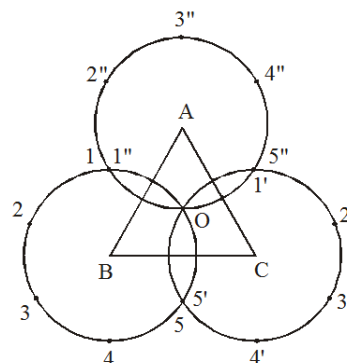
$$= \frac{(4 + 3\sqrt{2})}{2} g\ell$$

8. Ans (12)

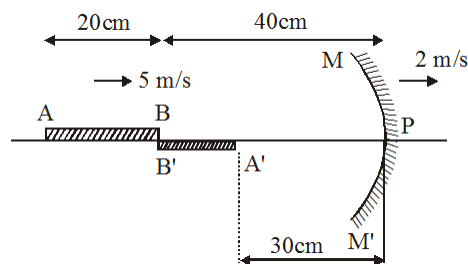
Images formed due to multiple reflection in mirrors AB and BC will lie on a circle centred at B. Position of images are 1, 2, 3, 4, 5. Images due to reflections in BC and CA are 1', 2', 3', 4' and 5'. Images due to reflections in AB and AC are 1'', 2'', 3'', 4'' and 5''.

1 and 1'', 1' and 5'' and 5 and 5' coincide.

$\therefore$ Total number of images = 12.



9. Ans (9)



$$\frac{1}{v_A} + \frac{1}{-60} = \frac{1}{-20}$$

$$\frac{1}{v_A} = \frac{1}{60} = \frac{1}{20}; \Rightarrow v_A = -30 \text{ cm}$$

$$V_{Im} = -\left(\frac{v}{u}\right)^2 \cdot V_{Om}$$

[V_{Im} = velocity of image wrt mirror, V_{Om} = velocity of object wrt mirror]

$$V_B' - V_m = -\left(\frac{40}{40}\right)^2 \cdot (V_B - V_m)$$

$$V_B' - 2 = -[5 - 2]$$

$$V_B' = -1 \text{ m/s} = 1 \text{ m/s} (\rightarrow)$$

A' (Image of A)

$$V_A' - V_m = -\left(\frac{30}{60}\right)^2 \cdot (V_A - V_m)$$

$$V_A' - 2 = -\frac{1}{4}[5 - 2]$$

$$V_A' = +\frac{5}{4} \text{ m/s} = \frac{5}{4} \text{ m/s} (\rightarrow)$$

Length A'B' is increasing at a rate of

$$1 + \frac{5}{4} = \frac{9}{4} \text{ m/s}$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (C)**

$$\therefore (1 + \sqrt{2})^n = A_n + B_n \sqrt{2}$$

$$\Rightarrow (1 - \sqrt{2})^n = A_n - B_n \sqrt{2}$$

$$\text{on solving, } A_n = \frac{(1 + \sqrt{2})^n + (1 - \sqrt{2})^n}{2}$$

$$B_n = \frac{(1 + \sqrt{2})^n - (1 - \sqrt{2})^n}{2\sqrt{2}}$$

$\therefore$

$$\lim_{n \rightarrow \infty} \frac{A_n}{B_n} = \lim_{n \rightarrow \infty} \sqrt{2} \cdot \left(\frac{1 + \left(\frac{1 - \sqrt{2}}{1 + \sqrt{2}} \right)^n}{1 - \left(\frac{1 - \sqrt{2}}{1 + \sqrt{2}} \right)^n} \right) = \sqrt{2}$$

$$\therefore \left(\frac{1 - \sqrt{2}}{1 + \sqrt{2}} \right)^n \rightarrow 0 \text{ as } n \rightarrow \infty$$

$$\& (A) \rightarrow \text{limit} = 1 \quad (C) \rightarrow \text{limit} = \sqrt{2}$$

$$(B) \rightarrow \text{limit} = 0$$

2. **Ans (C)**

$$\lim_{x \rightarrow -\infty} \sqrt{x^2 - \frac{4x}{3}} + x + 2$$

$$\lim_{x \rightarrow -\infty} \left(\frac{x^2 - \frac{4x}{3} - (x + 2)^2}{\sqrt{x^2 - \frac{4x}{3}} - (x + 2)} \right)$$

$$\lim_{x \rightarrow -\infty} \frac{-\frac{16}{3}x - 4}{|x| \sqrt{1 - \frac{4}{3x}} - (x + 2)}$$

$$\lim_{x \rightarrow -\infty} \frac{x \left(-\frac{16}{3} - \frac{4}{x} \right)}{x \left[-\sqrt{1 - \frac{4}{3x}} - \left(1 + \frac{2}{x} \right) \right]}$$

$$\lim_{x \rightarrow -\infty} -\frac{16}{3(-2)} = \frac{8}{3}$$

3. **Ans (A,B)**

$$\text{Clearly } \lim_{x \rightarrow \alpha^+} \frac{\sin([x^3] - [x]^3)}{(x - \alpha)^3} = 0$$

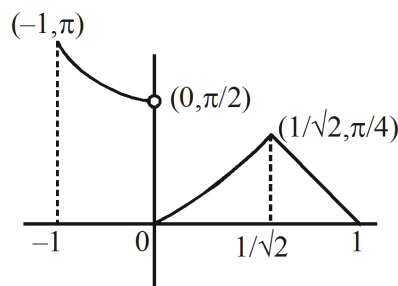
$$\therefore \text{for existence of limit L.H.L} = 0.$$

$$\Rightarrow \alpha^3 - 1 - (\alpha - 1)^3 = 0$$

$$\Rightarrow \alpha^3 - 1 - (\alpha^3 - 3\alpha^2 + 3\alpha - 1) = 0$$

$$3\alpha^2 - 3\alpha = 0 \Rightarrow \alpha = 0, 1$$

4. **Ans (A,C,D)**



$$\lambda_1 = 2, \lambda_2 = 2, \lambda_3 = 1$$

5. **Ans (A,B,C,D)**

Doyourself

6. **Ans (A,B,C)**

$$-1 \leq \sum_{k=1}^n 3^k (f(x + ky) - f(x - ky)) \leq 1$$

$$\text{Also, } -1 \leq \sum_{k=1}^{n-1} 3^k (f(x + ky) - f(x - ky)) \leq 1$$

$$-2 \leq 3^n (f(x + ny) - f(x - ny)) \leq 2$$

$$\frac{-2}{3^n} \leq (f(x + ny) - f(x - ny)) \leq \frac{2}{3^n}$$

$$\Rightarrow f(x + ny) - f(x - ny) = 0$$

$$\Rightarrow f(x) \text{ is constant function}$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. **Ans (D)**

$$\tan^2 \alpha \tan^2 \beta + \tan^2 \beta \tan^2 \gamma + \tan^2 \gamma \tan^2 \alpha + 2 \tan^2 \alpha \tan^2 \beta \tan^2 \gamma = 1.$$

on dividing equation by $\tan^2 \alpha \tan^2 \beta \tan^2 \gamma$, we get

$$\cot^2 \gamma + \cot^2 \alpha + \cot^2 \beta + 2 = \cot^2 \alpha \cot^2 \beta \cot^2 \gamma$$

$$\Pi \cot^2 \alpha - \Sigma \cot^2 \alpha = 2$$

8. **Ans (C)**

$$\text{Put } \tan^2 \alpha = x$$

$$\tan^2 \beta = y$$

$$\tan^2 \gamma = z$$

$$\text{given condition } xy + yz + zx + 2xyz = 1 \dots (i)$$

$$\text{Now } \cos 2\alpha + \cos 2\beta + \cos 2\gamma$$

$$= \frac{1-x}{1+x} + \frac{1-y}{1+y} + \frac{1-z}{1+z}$$

$$= \frac{3+x+y+z-xy-yz-zx-3xyz}{1+x+y+z+xy+yz+zx+xyz}$$

Using (i), we get

$$= \frac{2+x+y+z-xyz}{2+x+y+z-xyz} = 1$$

9. Ans (D)

$$\text{Given } a_{n+1} = a_n + 4n + 3$$

Now

$$\begin{aligned} a_k - a_1 &= \sum_{n=1}^{k-1} (a_{n+1} - a_n) = \sum_{n=1}^{k-1} (4n + 3) \\ &= 4(k-1) \frac{(k)}{2} + 3(k-1) = 2k^2 + k - 3 \\ \Rightarrow a_k &= 2k^2 + k - 3 \end{aligned}$$

So,

$$\begin{aligned} \lim_{k \rightarrow \infty} \frac{\sqrt{2k^2 + k - 3} + \sqrt{2(4k)^2 + (4k) - 3} + \dots + \sqrt{2(4^{10}k)^2 + (4^{10}k) - 3}}{\sqrt{2k^2 + k - 3} + \sqrt{2(2k)^2 + (2k) - 3} + \dots + \sqrt{2(2^{10}k)^2 + (2^{10}k) - 3}} \\ = \lim_{k \rightarrow \infty} \frac{\sqrt{2 + \frac{1}{k} - \frac{3}{k^2}} + \sqrt{2(4)^2 + \frac{4}{k} - \frac{3}{k^2}} + \dots + \sqrt{2(4^{10})^2 + \frac{4^{10}}{k} - \frac{3}{k^2}}}{\sqrt{2 + \frac{1}{k} - \frac{3}{k^2}} + \sqrt{2(2)^2 + \frac{2}{k} - \frac{3}{k^2}} + \dots + \sqrt{2(2^{10})^2 + \frac{2^{10}}{k} - \frac{3}{k^2}}} \\ = \frac{\sqrt{2} + \sqrt{2(4^2)} + \dots + \sqrt{2(4^{10})^2}}{\sqrt{2} + \sqrt{2(2^2)} + \dots + \sqrt{2(2^{10})^2}} \\ = \frac{\sqrt{2}(1 + 4 + 4^2 + \dots + 4^{10})}{\sqrt{2}(1 + 2 + 2^2 + \dots + 2^{10})} = \frac{\frac{1}{3}(4^{11} - 1)}{2^{11} - 1} = 683 \end{aligned}$$

10. Ans (B)

$$\text{Given } a_{n+1} = a_n + 4n + 3$$

Now

$$\begin{aligned} a_k - a_1 &= \sum_{n=1}^{k-1} (a_{n+1} - a_n) = \sum_{n=1}^{k-1} (4n + 3) \\ &= 4(k-1) \frac{(k)}{2} + 3(k-1) = 2k^2 + k - 3 \\ \Rightarrow a_k &= 2k^2 + k - 3 \end{aligned}$$

So,

$$\begin{aligned} \lim_{k \rightarrow \infty} \frac{\sqrt{2k^2 + k - 3} + \sqrt{2(4k)^2 + (4k) - 3} + \dots + \sqrt{2(4^{10}k)^2 + (4^{10}k) - 3}}{\sqrt{2k^2 + k - 3} + \sqrt{2(2k)^2 + (2k) - 3} + \dots + \sqrt{2(2^{10}k)^2 + (2^{10}k) - 3}} \\ = \lim_{k \rightarrow \infty} \frac{\sqrt{2 + \frac{1}{k} - \frac{3}{k^2}} + \sqrt{2(4)^2 + \frac{4}{k} - \frac{3}{k^2}} + \dots + \sqrt{2(4^{10})^2 + \frac{4^{10}}{k} - \frac{3}{k^2}}}{\sqrt{2 + \frac{1}{k} - \frac{3}{k^2}} + \sqrt{2(2)^2 + \frac{2}{k} - \frac{3}{k^2}} + \dots + \sqrt{2(2^{10})^2 + \frac{2^{10}}{k} - \frac{3}{k^2}}} \\ = \frac{\sqrt{2} + \sqrt{2(4^2)} + \dots + \sqrt{2(4^{10})^2}}{\sqrt{2} + \sqrt{2(2^2)} + \dots + \sqrt{2(2^{10})^2}} \\ = \frac{\sqrt{2}(1 + 4 + 4^2 + \dots + 4^{10})}{\sqrt{2}(1 + 2 + 2^2 + \dots + 2^{10})} = \frac{\frac{1}{3}(4^{11} - 1)}{2^{11} - 1} = 683 \end{aligned}$$

PART-3 : MATHEMATICS

SECTION-II (i)

1. Ans (3.00)

$$f(x) = \tan^{-1} \left(\frac{|x|}{\sqrt{1-x^2}} \right) + \cos^{-1} \left(\frac{1}{\sqrt{1+x^2}} \right)$$

$$\text{Put } |x| = \sin \theta \quad \text{Put } |x| = \tan \theta$$

$$\theta \in \left(0, \frac{\pi}{2}\right) \quad \theta \in \left(0, \frac{\pi}{2}\right)$$

$$\therefore f(x) = \sin^{-1} |x| + \tan^{-1} |x|$$

Since $\sin^{-1} |x| + \tan^{-1} |x|$ are both increasing function in $[0, 1]$.

$$\therefore f(x) \in \left[0, \frac{3\pi}{4}\right]$$

Integer's in range = $\{0, 1, 2\}$

$$f(\sin \theta) > f(\cos \theta)$$

$$\therefore \sin \theta > \cos \theta$$

$$\theta \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$$

2. Ans (0.00)

$$f(x) = \tan^{-1} \left(\frac{|x|}{\sqrt{1-x^2}} \right) + \cos^{-1} \left(\frac{1}{\sqrt{1+x^2}} \right)$$

$$\text{Put } |x| = \sin \theta \quad \text{Put } |x| = \tan \theta$$

$$\theta \in \left(0, \frac{\pi}{2}\right) \quad \theta \in \left(0, \frac{\pi}{2}\right)$$

$$\therefore f(x) = \sin^{-1} |x| + \tan^{-1} |x|$$

Since $\sin^{-1} |x| + \tan^{-1} |x|$ are both increasing function in $[0, 1]$.

$$\therefore f(x) \in \left[0, \frac{3\pi}{4}\right]$$

Integer's in range = $\{0, 1, 2\}$

$$f(\sin \theta) > f(\cos \theta)$$

$$\therefore \sin \theta > \cos \theta$$

$$\theta \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$$

3. **Ans (0.75)**

$$\begin{aligned} a + b - c &= \sin 10^\circ + \sin 50^\circ - \sin 70^\circ \\ &= \sin 10^\circ + 2\cos 60^\circ \sin(-10^\circ) = 0 \\ \therefore a + b &= c \\ \therefore 8abc \left(\frac{a+b}{c} \right) \left(\frac{bc+ac-ab}{abc} \right) \\ 8(c^2 - ab) &= 8[\sin^2 70^\circ - \sin 10^\circ \sin 50^\circ] \\ &= 4[2\sin^2 70^\circ - \cos 40^\circ + \cos 60^\circ] \\ &= 4[1 + \cos 40^\circ - \cos 40^\circ + \cos 60^\circ] \\ &= 4 \left[\frac{3}{2} \right] = 6 \end{aligned}$$

$$\begin{aligned} (a + b - c) &= 0 \\ a^2 + b^2 + c^2 + 2(ab - bc - ca) &= 0 \\ a^4 + b^4 + c^4 + 2(a^2b^2 + b^2c^2 + c^2a^2) \\ &= 4[(ab)^2 + (bc)^2 + (ca)^2] = 2 \\ \frac{a^4 + b^4 + c^4}{(ab)^2 + (bc)^2 + (ca)^2} &= 2 \end{aligned}$$

4. **Ans (2.00)**

$$\begin{aligned} a + b - c &= \sin 10^\circ + \sin 50^\circ - \sin 70^\circ \\ &= \sin 10^\circ + 2\cos 60^\circ \sin(-10^\circ) = 0 \\ \therefore a + b &= c \\ \therefore 8abc \left(\frac{a+b}{c} \right) \left(\frac{bc+ac-ab}{abc} \right) \\ 8(c^2 - ab) &= 8[\sin^2 70^\circ - \sin 10^\circ \sin 50^\circ] \\ &= 4[2\sin^2 70^\circ - \cos 40^\circ + \cos 60^\circ] \\ &= 4[1 + \cos 40^\circ - \cos 40^\circ + \cos 60^\circ] \\ &= 4 \left[\frac{3}{2} \right] = 6 \end{aligned}$$

$$\begin{aligned} (a + b - c) &= 0 \\ a^2 + b^2 + c^2 + 2(ab - bc - ca) &= 0 \\ a^4 + b^4 + c^4 + 2(a^2b^2 + b^2c^2 + c^2a^2) \\ &= 4[(ab)^2 + (bc)^2 + (ca)^2] = 2 \\ \frac{a^4 + b^4 + c^4}{(ab)^2 + (bc)^2 + (ca)^2} &= 2 \end{aligned}$$

5. **Ans (7.00)**

$$\begin{aligned} 9 - (x-1)(x-3)(x-5)(x-7) &\geq 0 \\ 9 - (x^2 - 8x + 7)(x^2 - 8x + 15) &\geq 0 \\ \text{Let } x^2 - 8x + 11 = t, \text{ where } t \text{ can be in } [-5, \infty) \\ \dots\dots\dots(1) \\ \Rightarrow 9 - (t-4)(t+4) &\geq 0 \\ \Rightarrow 25 - t^2 \geq 0 \Rightarrow t \in [-5, 5] \\ \Rightarrow -5 \leq x^2 - 8x + 11 \leq 5 \\ \Rightarrow x^2 - 8x + 16 \geq 0 \text{ and } x^2 - 8x + 6 \leq 0 \\ \Rightarrow x \in \mathbb{R} \text{ and } x \in [4 - \sqrt{10}, 4 + \sqrt{10}] \end{aligned}$$

So integral values of $x \in \{1, 2, 3, 4, 5, 6, 7\}$

6. **Ans (2.00)**

$$\begin{aligned} f(x) &= \sqrt{9 - (x-1)(x-3)(x-5)(x-7)} \\ &= \sqrt{25 - t^2}, \text{ where } t^2 \in [0, 25] \\ \Rightarrow f(x) &\in [0, 5], \text{ obvious many-one (think)} \end{aligned}$$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. **Ans (561)**

$$\begin{aligned} f(94) &= 94^2 - f(93) \\ &= 94^2 - 93^2 + f(92) \\ &\vdots \\ &= (94^2 - 93^2) + (92^2 - 91^2) + \dots + (22^2 - 21^2) \\ &\quad + 20^2 - f(19) \\ f(94) &= 4561 \end{aligned}$$

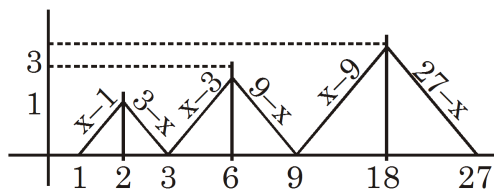
8. **Ans (8)**

$$\begin{aligned} \lim_{x \rightarrow 0} \left(\frac{f(x) - x + 2x^2 - 3x^3}{3x^3} \right) \cdot \frac{2}{x} &= 4 \\ \Rightarrow f(x) &= x - 2x^2 + 3x^3 + 6x^4 \\ \Rightarrow \sum_{i=0}^4 a_i &= 1 - 2 + 3 + 6 = 8 \end{aligned}$$

9. **Ans (429)**

$$f(x) = \begin{cases} x - 3^n; & 3^n \leq x \leq 2 \times 3^n \\ 3^{(n+1)} - x; & 2 \times 3^n < x \leq 3^{(n+1)} \end{cases}; n \in \mathbb{N} \cup \{0\}$$

Graph of function



$$\begin{aligned} \text{Since } 2 \times 3^6 &< 2001 < 3^7 \\ \Rightarrow f(2001) &= 3^7 - 2001 = 186 \\ \text{Now, } f(x) &\leq 81, \text{ when } x \leq 3^5 \\ \text{And } 0 \leq f(x) &\leq 3^5 = 243 \\ \text{when } 3^5 &\leq x \leq 2 \times 3^5 \\ \text{So, } 186 &= x - 3^5 \\ \Rightarrow x &= 243 + 186 = 429 \end{aligned}$$